

NAINSY SHARMA

Aspiring Data & AI Professional

Raipur, India | nainsysharma24@gmail.com | [LinkedIn](#) | [GitHub](#)

SUMMARY

Final-year B.Tech (CSE - AI & ML) student with hands-on experience across SQL, Python, and machine learning to solve real business problems - from credit risk and churn prediction to threat detection - and turn model output into deployable, decision-ready dashboards. Combines technical execution with structured business problem framing and clear stakeholder communication, developed through case-solving and guesstimate practice.

EDUCATION

Vellore Institute of Technology (VIT)

B.Tech, Computer Science & Engineering (AI & Machine Learning) | CGPA: 8.42 / 10 | 2023 - 2027

SKILLS

Programming & Querying: Python (Pandas, NumPy), SQL (PostgreSQL, window functions), Excel

Machine Learning: scikit-learn, XGBoost, classification, clustering (K-Means), RFM analysis, feature engineering

Visualization & BI: Power BI-ready analysis, Plotly, Matplotlib, Seaborn, Streamlit dashboards

Tools & Deployment: FastAPI, Git/GitHub, Jupyter, REST APIs, Render, Streamlit Cloud

AI-Assisted Development: Claude, ChatGPT, Gemini - used to accelerate coding, debugging, feature engineering, and documentation across projects

Core Strengths: Business problem framing, guesstimates & case-solving, stakeholder communication, data storytelling

CERTIFICATIONS

- **Applied Machine Learning in Python** — University of Michigan (Coursera)
- **Data Science Job Simulation** — BCG (Forage)

PROJECTS

RiskLens - Credit Risk Analysis & Loan Default Prediction | Python, SQL, XGBoost, FastAPI, Streamlit

[GitHub Repo](#) · [Live Dashboard](#) · 2025

- Analyzed ~100,000 Lending Club loans using SQL window functions to compute default rates by grade, purpose, and vintage cohort, finding default risk ranged from ~5% (**Grade A**) to ~59% (**Grade G**)
- Built and benchmarked Logistic Regression, Random Forest, and XGBoost models; identified and removed data leakage that had inflated ROC-AUC from 0.96 to a realistic **0.74**, improving model trustworthiness
- Deployed a production-style microservice (FastAPI backend + Streamlit frontend) delivering **live default-risk predictions** through an interactive, publicly accessible dashboard

ChurnIQ - AI-Powered Customer Intelligence & Churn Prediction | Python, scikit-learn, XGBoost, RFM, K-Means

[GitHub Repo](#) · [Live Dashboard](#) · 2025

- Engineered 40 features (12 derived) on a 5,000-customer telecom dataset and trained 4 ML models; best model (Logistic Regression) achieved **0.763 ROC-AUC** and **0.717 F1-score**
- Applied RFM analysis and K-Means clustering to segment customers into 4 risk tiers, flagging **963 critical-risk customers** and **\$1.88M in annual revenue at risk**
- Translated model output into a prioritized retention action plan (targeted discounts, contract upgrades, loyalty offers) projected to save **~\$527K annually**, presented via an interactive dashboard

Multi-Domain Threat Detection System | Python, scikit-learn, Streamlit, Plotly

[GitHub Repo](#) · [Live Dashboard](#) · 2025

- Built a real-time classification system across Air, Land, and Water sensor domains on a 525-record, 34-feature dataset, comparing Decision Tree, Random Forest, and KNN models
- Converted raw multi-sensor inputs (velocity, altitude, trajectory, radar/thermal signatures) into **HIGH / MEDIUM / LOW** threat classifications with confidence scoring for decision support

LEADERSHIP & EXTRA-CURRICULAR

- **Core Member, National Service Scheme (NSS)**, VIT Bhopal - organized community outreach and awareness initiatives
- **Member, Student Council & Discipline Committee** - co-organized VIT Bhopal's annual college fest, coordinating logistics across multiple student teams
- State-Level Sprinter (100m & 200m) and competitive **Basketball** player - balanced high-performance athletics with academics, reflecting discipline and consistency